



Stock Assessment of Yellowfin Tuna in the WCPO



A. Magnusson, T. Teears, K. Kim, J. Hampton, N. Davies,
J. Scutt Phillips, C. Castillo Jordán, T. Peatman, P. Hamer

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Objectives in the stock assessment

- ▶ Deliver report **on time**
- ▶ Develop a model with **good convergence properties**
- ▶ **Simplify** the model as much as possible
- ▶ Incorporate **innovations** recommended by experts, including the SC
- ▶ Check for and **remove errors**

PAW 2026 discussion points

Spatial structure

1. Consider single-region or equatorial model

Fisheries

2. Convert weight comps to length comps
3. Combine fisheries
4. Dirichlet-multinomial sample sizes

Recruitment

5. Orthogonal polynomial recruitment

Natural mortality

6. Fix M at different levels

Movement

7. SEAPODYM movement coefficients

Growth

8. Reweight otoliths based on location
9. Richards growth curve

Selectivity

10. Length-based selectivity

CPUE

11. Regional CPUE analysis w/thermal limits
12. Regional scaling with a prior
13. Time-varying CV
14. Effort creep

Tags

15. Mixing period for each release group

New features in the assessment

- ✓ Consider single-region or equatorial model
- ✓ Convert weight comps to length comps
- ✓ Combine fisheries
 - Dirichlet-multinomial sample sizes
- ✓ Orthogonal polynomial recruitment
- ✓ Fix M at different levels
 - SEAPODYM movement coefficients
- ✓ Reweight otoliths based on location
 - Richards growth curve
 - Length-based selectivity
- ✓ Regional CPUE analysis w/thermal limits
- ✓ Regional scaling with a prior
- ✓ Time-varying CV
- ✓ Effort creep
 - Mixing period for each release group
 - + Tag rep. rates excl. during mixing period
 - + Tag variance with overdispersion $\tau=2$

Model overview

MULTIFAN-CL version 2.2.7.9

5 regions

1952–2024, quarterly time steps

31 fisheries + 5 CPUE indices

Fixed M at terminal age, Lorenzen shape

Internally estimated growth, von Bertalanffy

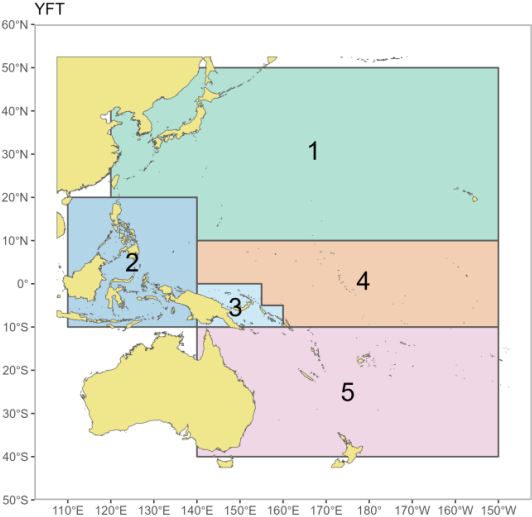
Major reduction in parameters 1901 → 454 (was 11,688 in 2020 assessment)

Regional scaling with multivariate normal prior

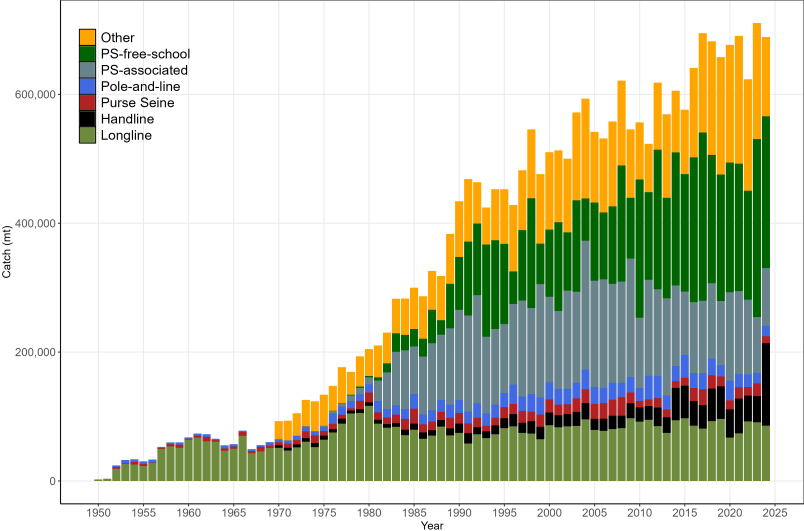
Likelihoods based on model fits to data

- Length comps - CAAL (otoliths)
- CPUE - Tags

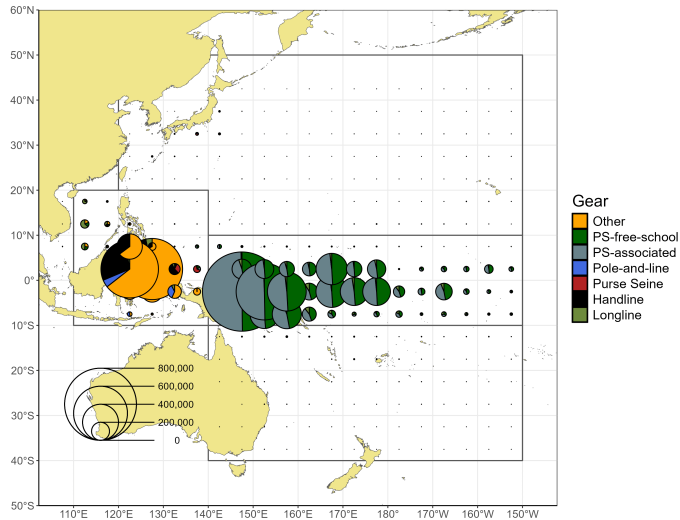
Regions



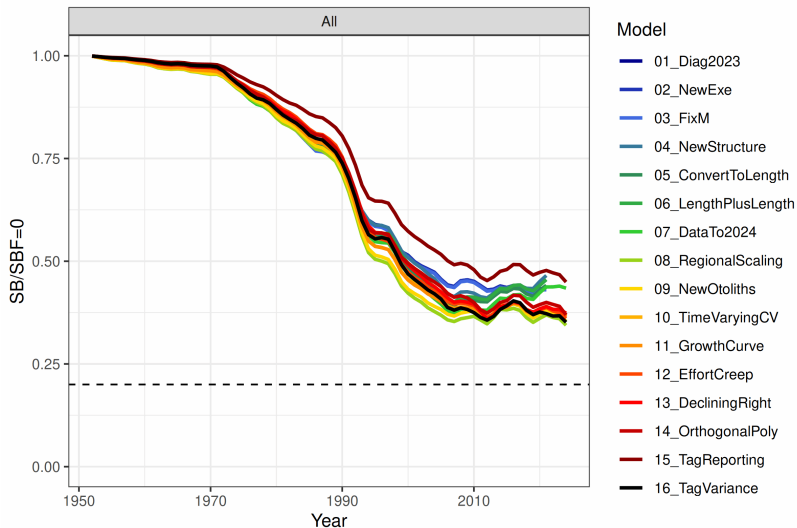
Catches



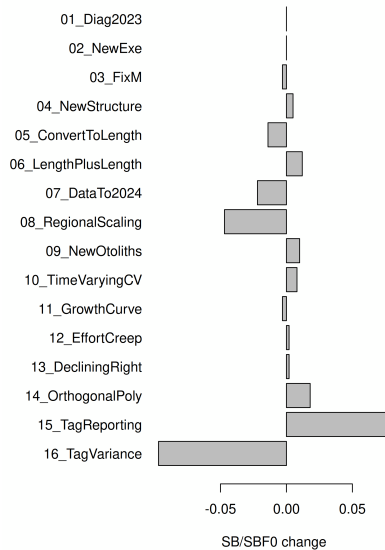
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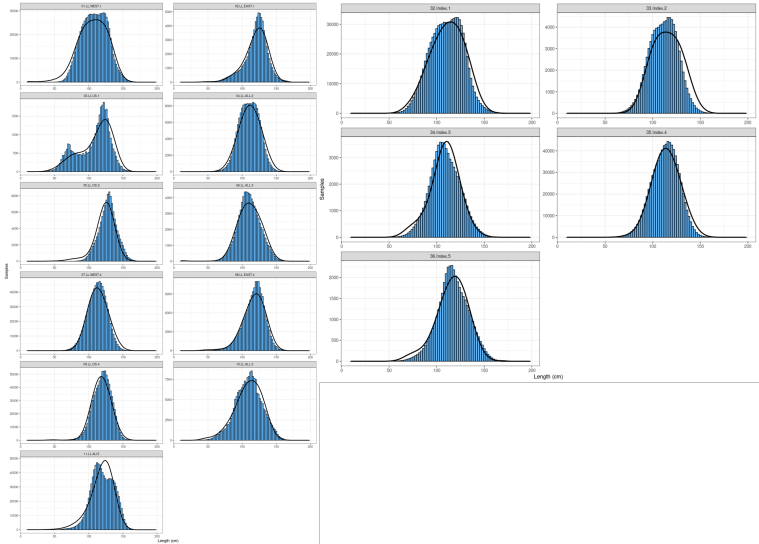
Stepwise



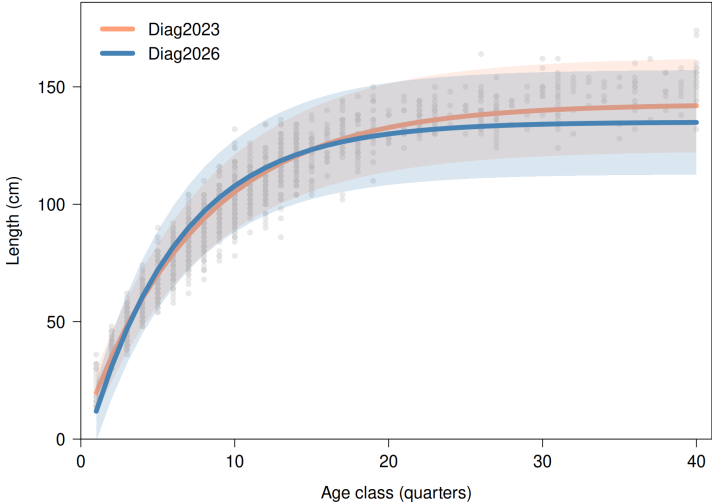
Stepwise



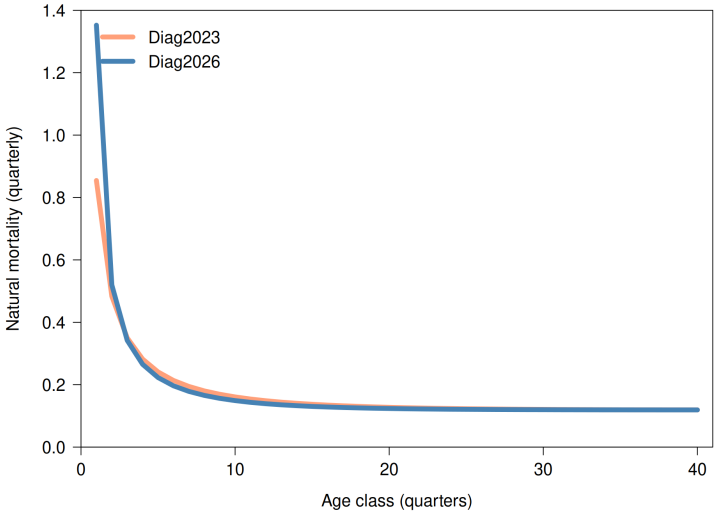
Fit to longline and index fisheries



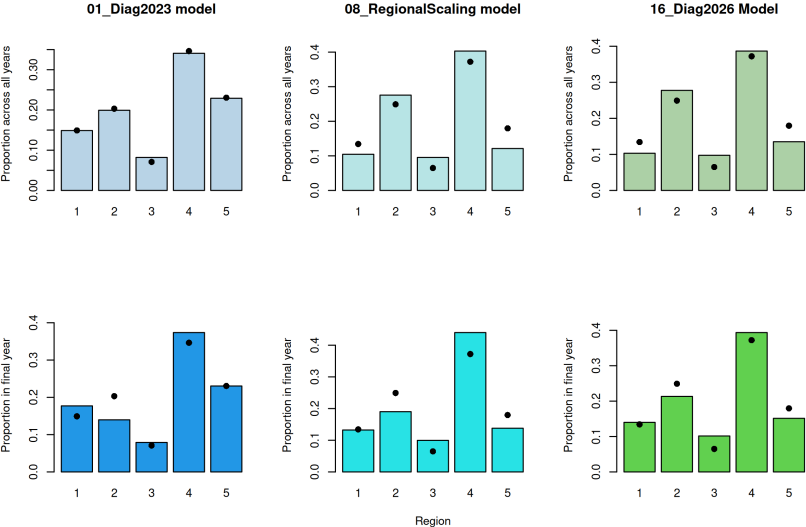
Growth



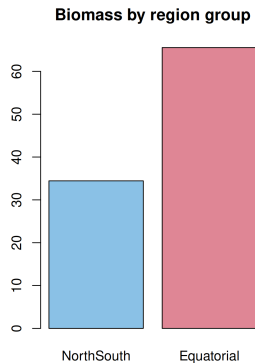
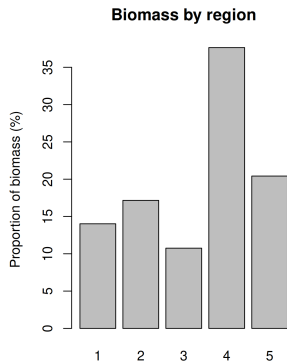
Natural mortality



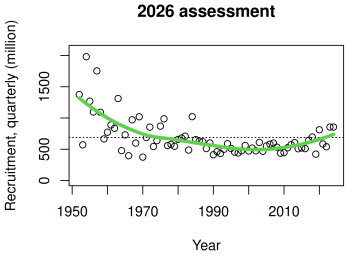
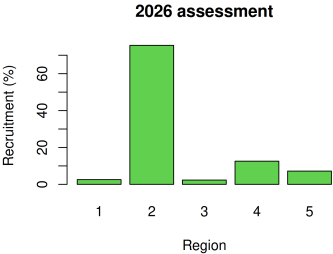
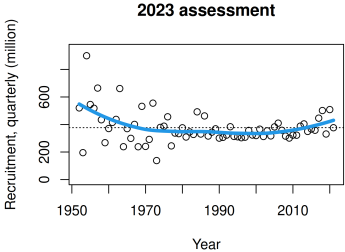
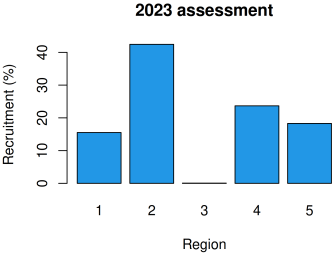
Regional scaling



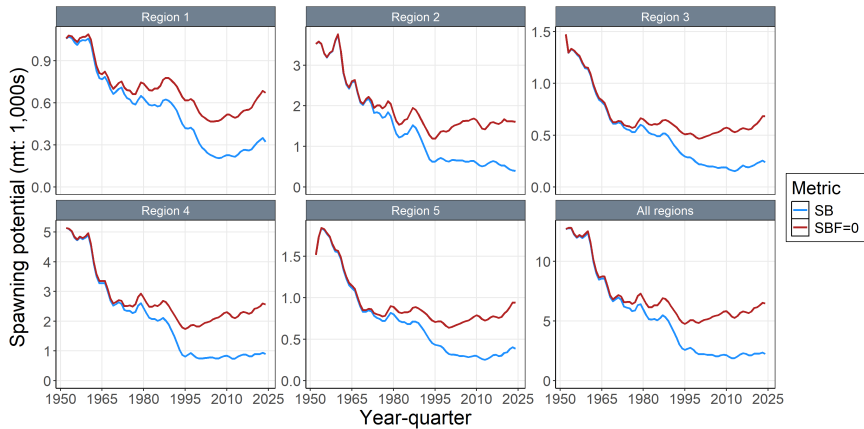
Biomass distribution between regions



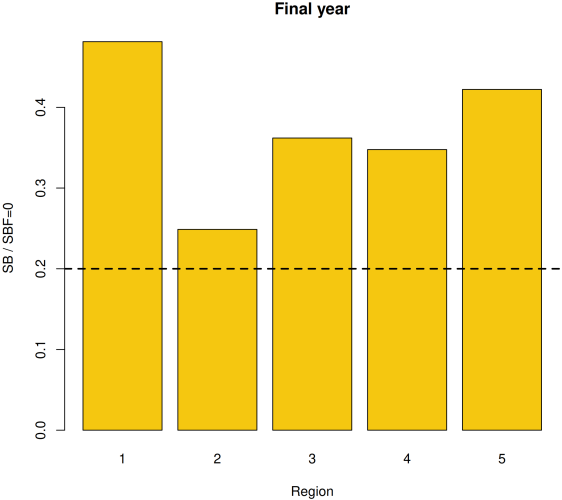
Recruitment



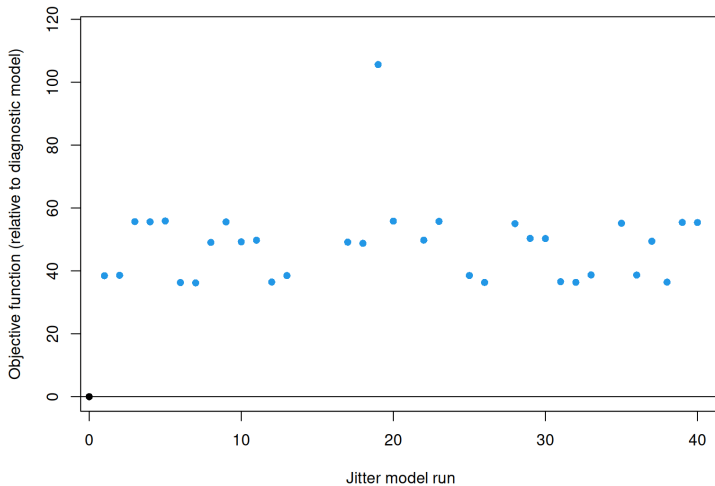
Fished and unfished biomass by region



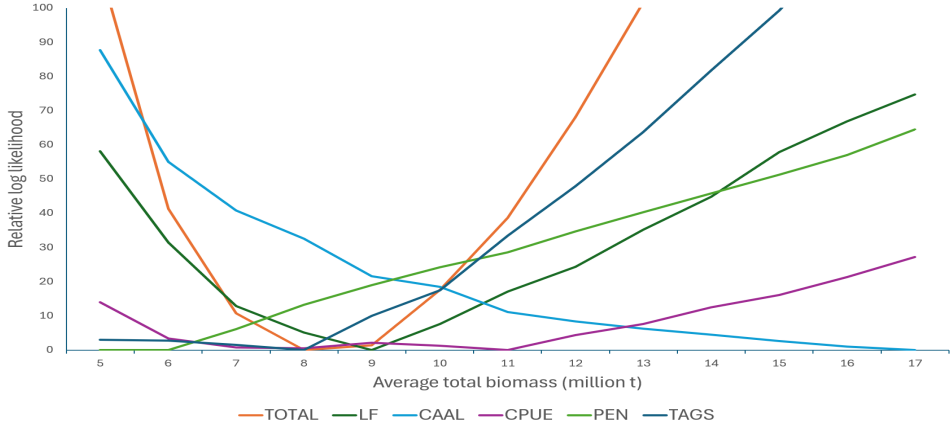
Depletion



Jitter analysis



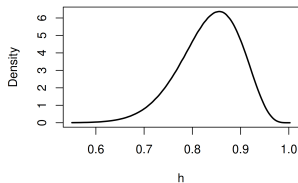
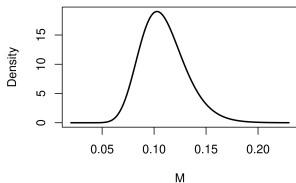
Likelihood profile



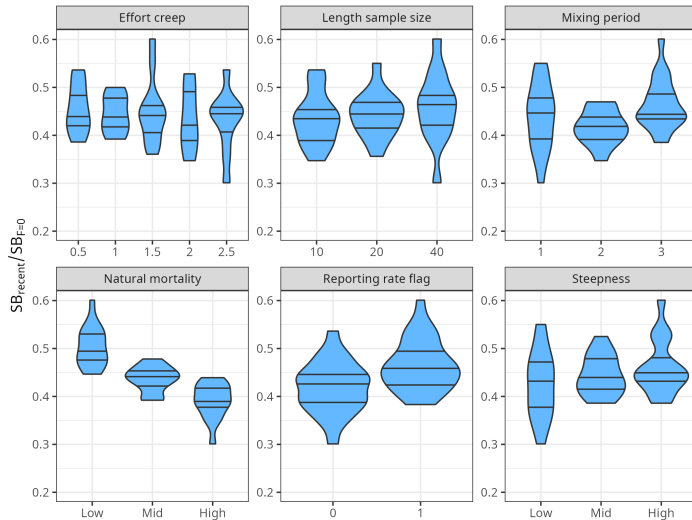
Ensemble analysis

Structural uncertainty

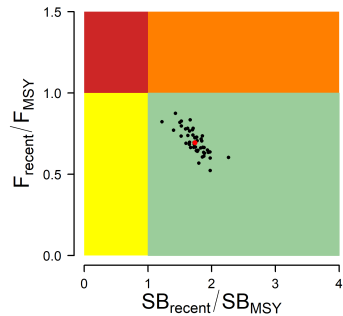
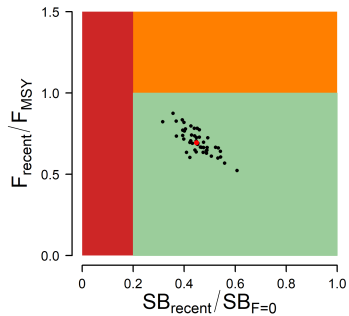
- ▶ Effort creep: 0.5, 1.0, 1.5, 2.0, 2.5% for the first years, half of that in later years
- ▶ Length sample size: divided by 10, 20, 40
- ▶ Tag mixing period: 1, 2, 3
- ▶ Tag reporting rate during mixing period: 0=included, 1=excluded
- ▶ Natural mortality: continuous prior
- ▶ Steepness: continuous prior



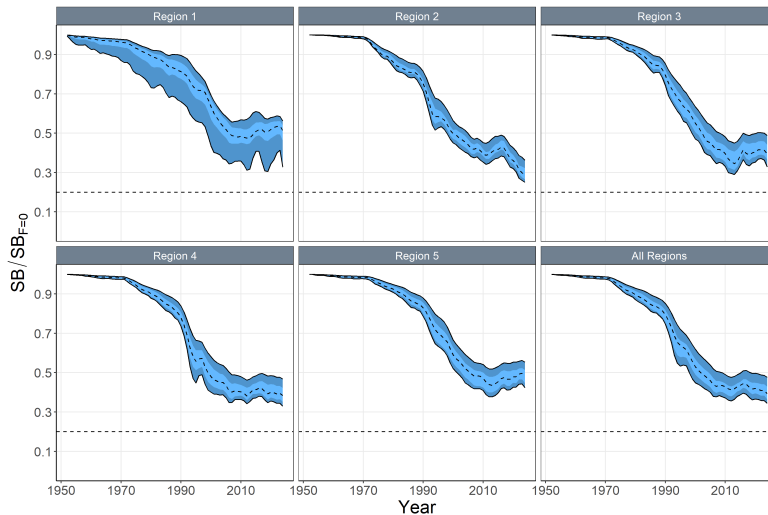
Ensemble analysis



Stock status



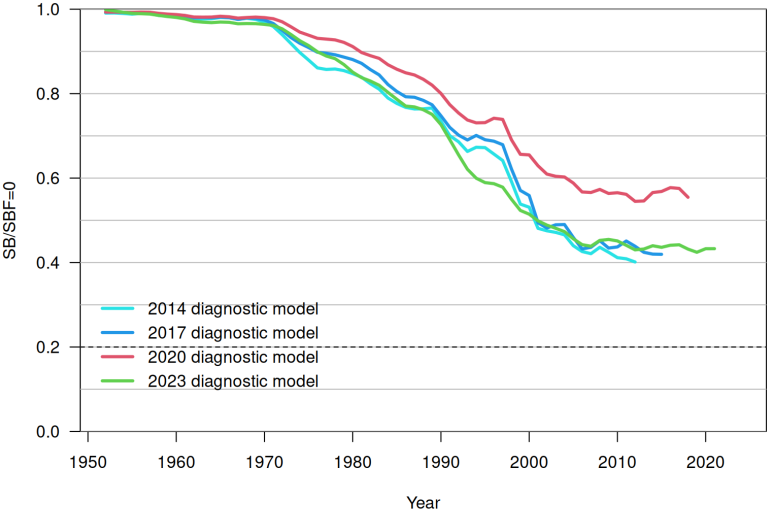
Stock status



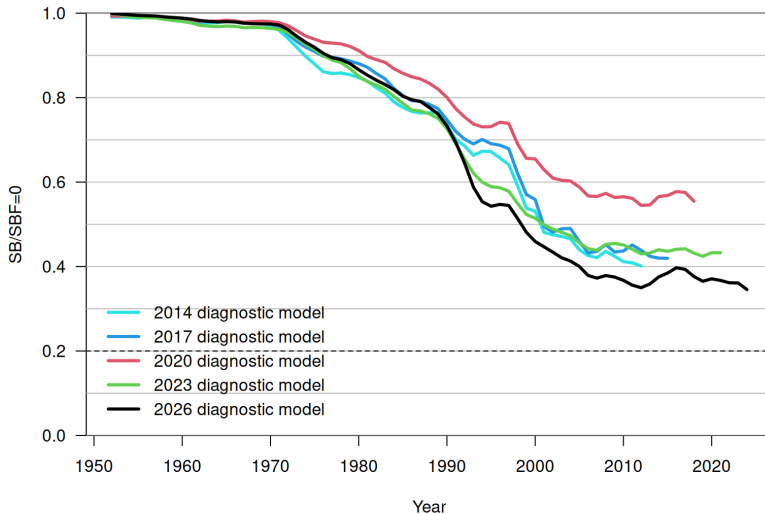
Stock status

Quantity	Mean	Median	Min	10%	90%	Max
F_{MSY}	0.05	0.05	0.03	0.03	0.07	0.09
f_{mult}	1.44	1.44	1.14	1.24	1.64	1.91
$F_{\text{recent}}/F_{\text{MSY}}$	0.70	0.69	0.52	0.61	0.80	0.87
MSY	709,492	712,800	610,000	657,080	758,040	838,800
SB_{latest}	4,225,501	3,601,588	1,449,490	1,781,367	7,185,084	12,284,825
SB_{recent}	4,239,072	3,595,731	1,546,934	1,845,127	7,163,028	12,139,288
$SB_{F=0}$	9,116,300	8,293,368	4,138,987	5,214,403	13,617,615	20,203,280
$SB_{\text{latest}}/SB_{F=0}$	0.44	0.44	0.28	0.37	0.53	0.61
$SB_{\text{latest}}/SB_{\text{MSY}}$	1.72	1.73	1.11	1.49	1.89	2.10
SB_{MSY}	2,409,656	2,076,000	758,100	1,246,700	3,799,600	6,074,000
$SB_{\text{MSY}}/SB_{F=0}$	0.26	0.25	0.18	0.23	0.29	0.30
$SB_{\text{recent}}/SB_{F=0}$	0.44	0.44	0.30	0.38	0.53	0.60
$SB_{\text{recent}}/SB_{\text{MSY}}$	1.74	1.75	1.18	1.51	1.91	2.22
$Y_{F_{\text{current}}}$	167,600	166,700	150,500	158,050	178,030	192,400

Comparison with previous assessments



Comparison with previous assessments



Main conclusions

- ▶ Spawning depletion has been relatively stable since around 2005 to the present, between 0.35 and 0.50
- ▶ All the ensemble models were above the $SB_{\text{current}}/SB_{F=0} = 0.20$ limit reference point and $F_{\text{current}} < F_{\text{MSY}}$
- ▶ Using a prior for M in the ensemble analysis improved the incorporation of uncertainty about this key parameter in the assessment
- ▶ Convergence problems remain in the model, likelihood conflict between the data sources
- ▶ Slightly lower estimated relative stock size compared to previous assessment, primarily because of model settings related to tags
- ▶ Orthogonal polynomial recruitment reduced the number of estimated parameters and improved model convergence
- ▶ Effort creep incorporated in the assessment for the first time

Recommendations for further work

1. Examine and describe data conflict in greater detail
2. Consider alternative regional structure
3. Explore how the growth curve is affected by the tagging data likelihood
4. Workshop that focuses on how best to analyze the tagging data for the assessment

Browse results online

Shiny app with stepwise results:

<https://ofp-sam.shinyapps.io/yft-2026-stepwise>

After SC:

<https://github.com/PacificCommunity/ofp-sam-yft-2026-diagnostic>

<https://github.com/PacificCommunity/ofp-sam-yft-2026-ensemble>